

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12397411
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Gottschling; Rafael et al.

Power tool stand

Abstract

A power tool stand for a removeable power tool has a first battery interface and a removeable battery having a first tool interface. The power tool stand comprises a base and a projecting frame connected to the base. A power tool carriage is moveably mounted on the projecting frame and the power tool carriage comprising a second battery interface and a second tool interface. The second battery interface is compatible with the first tool interface and the second battery interface is arranged to electrically and mechanically connect together with first tool interface. The second tool interface is compatible with the first battery interface and the second tool interface is arranged to electrically and mechanically connect together with first battery interface. The first battery interface and the first tool interface are incompatible.

Inventors:	Gottschling; Rafael (Selters-Eisenbach, DE), Kadlecek; Daniel (Bischofsheim, DE), Patel; Birju (Ruislip, GB)
Applicant:	BLACK & DECKER INC. (New Britain, CT)
Family ID:	1000008779057
Assignee:	Black & Decker Inc. (New Britain, CT)
Appl. No.:	18/687620
Filed (or PCT Filed):	September 06, 2022
PCT No.:	PCT/EP2022/074710
PCT Pub. No.:	WO2023/036766
PCT Pub. Date:	March 16, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240367308 A1	Nov. 07, 2024

Foreign Application Priority Data

GB

2112788

Sep. 08, 2021

Publication Classification

Int. Cl.: **B25H1/00** (20060101); **B23B39/10** (20060101); **B23B45/00** (20060101)

U.S. Cl.:

CPC **B25H1/0042** (20130101); **B23B39/10** (20130101); **B23B45/003** (20130101);
B25H1/0064 (20130101)

Field of Classification Search

CPC: B25H (1/0042); B25H (1/0064); B23B (39/10); B23B (45/003)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2466965	12/1948	Pitts	N/A	N/A
2629267	12/1952	Hart	N/A	N/A
3412813	12/1967	Johnson	N/A	N/A
3464655	12/1968	Schuman	N/A	N/A
3790059	12/1973	Jacke et al.	N/A	N/A
4365418	12/1981	Hoyss et al.	N/A	N/A
4375931	12/1982	Erdt	N/A	N/A
4500235	12/1984	Johnsen	N/A	N/A
4540149	12/1984	Rupprecht et al.	N/A	N/A
4634320	12/1986	Itzov	N/A	N/A
4655649	12/1986	Itzov et al.	N/A	N/A
5062743	12/1990	Wieland et al.	N/A	N/A
5713702	12/1997	Turner	N/A	N/A
5794724	12/1997	Moller	N/A	N/A
5899644	12/1998	Buck et al.	N/A	N/A
6223794	12/2000	Jones	N/A	N/A
6514018	12/2002	Martinez et al.	N/A	N/A
6692201	12/2003	Soderman	N/A	N/A
6792735	12/2003	Mohlenhoff	N/A	N/A
6981560	12/2005	Nguyen et al.	N/A	N/A
7214007	12/2006	Baratta	N/A	N/A
7347651	12/2007	Hintze et al.	N/A	N/A
7371034	12/2007	Clark	N/A	N/A
7476066	12/2008	Topf	N/A	N/A
7591615	12/2008	Li et al.	N/A	N/A
8337124	12/2011	Nguyen	N/A	N/A
9174283	12/2014	Timmons et al.	N/A	N/A
9452522	12/2015	Timmons et al.	N/A	N/A

9555481	12/2016	Nowland	N/A	N/A
9782865	12/2016	Richt et al.	N/A	N/A
9789600	12/2016	Jönsson et al.	N/A	N/A
10821525	12/2019	Carlson et al.	N/A	N/A
11148210	12/2020	Carlson et al.	N/A	N/A
11331730	12/2021	Carlson et al.	N/A	N/A
11858113	12/2023	Carlson et al.	N/A	N/A
11982184	12/2023	Schmitz	N/A	E21B 7/027
2008/0083106	12/2007	Elsworthy	N/A	N/A
2015/0290793	12/2014	Jönsson et al.	N/A	N/A
2016/0129582	12/2015	Ullrich	408/76	B23Q 3/1543
2016/0151873	12/2015	Richt et al.	N/A	N/A
2016/0151934	12/2015	Pfeifer et al.	N/A	N/A
2016/0250693	12/2015	Timmons et al.	N/A	N/A
2017/0232605	12/2016	Morton	N/A	N/A
2018/0119536	12/2017	Drexler et al.	N/A	N/A
2018/0133929	12/2017	Drexler et al.	N/A	N/A
2018/0297126	12/2017	Drexler et al.	N/A	N/A
2018/0297235	12/2017	Drexler et al.	N/A	N/A
2019/0093738	12/2018	Almqvist et al.	N/A	N/A
2019/0321896	12/2018	Carlson	N/A	B25H 1/0042
2021/0178571	12/2020	Denzel et al.	N/A	N/A
2023/0046097	12/2022	Hölderle et al.	N/A	N/A
2023/0070551	12/2022	Kugler et al.	N/A	N/A
2023/0234209	12/2022	Luo et al.	N/A	N/A
2024/0051096	12/2023	Sergyeyenko et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
20017256	12/1999	DE	N/A
20108167	12/2000	DE	N/A
10064173	12/2001	DE	N/A
10230460	12/2003	DE	N/A
202005020829	12/2005	DE	N/A
202005011822	12/2005	DE	N/A
202012003188	12/2011	DE	N/A
212018000029	12/2017	DE	N/A
1671769	12/2010	EP	N/A
3006145	12/2017	EP	N/A
3004517	12/2017	EP	N/A
4306267	12/2023	EP	N/A
622855	12/1948	GB	N/A
2299772	12/1995	GB	N/A
2018225379	12/2017	WO	N/A

OTHER PUBLICATIONS

International Search Report mailed Nov. 28, 2022 in PCT Application No. PCT/EP2022/074710, 16 pages. cited by applicant

International Search Report dated Dec. 12, 2022 in PCT Application No. PCT/EP2022/074708, 16 pages. cited by applicant

Notice of Allowance and Fees mailed Jul. 31, 2024 in U.S. Appl. No. 17/929,192, 8 pages. cited by applicant
Notice of Allowance and Fees mailed Dec. 18, 2024 in U.S. Appl. No. 17/929,172, 8 pages. cited by applicant
Non-Final Office Action mailed Dec. 18, 2024 in U.S. Appl. No. 17/929,161, 11 pages. cited by applicant
Extended European Search Report mailed Dec. 14, 2023 in EP Application No. 23177865.5, 8 pages. cited by applicant

Primary Examiner: Lopez; Michelle

Attorney, Agent or Firm: Womble Bond Dickinson (US) LLP

Background/Summary

RELATED APPLICATION

(1) This application is a national stage of PCT International Application No. PCT/EP2022/074710 filed Sep. 6, 2022, which claims priority to GB Patent Application No. 2112788.1 filed Sep. 8, 2021, the contents of both of which are incorporated herein by reference in their entirety.

TECHNICAL FIELD

(2) The present disclosure relates to a power tool stand. In particular the present disclosure relates to drill power tool stand.

BACKGROUND

(3) Some power tools are mountable on a tool stand. For example, certain drills can be mountable on a drill stand. This can improve the stability of the power tool during use and ensure that the power tool alignment with respect to the workpiece remains fixed during use. Furthermore, the power tool stand can be mountable onto the workpiece and this can further improve precision during operation.

(4) One such power tool stand is shown in US 2019/0321896 which shows a drill stand on which a battery powered drill can be mounted. A problem with this is that the drill power tool is very large and heavy and this makes mounting and using the drill power tool on the drill power tool stand cumbersome. For example, the base of the drill power tool stand may need to be weighted down during use to stop it toppling over when used in a vertical orientation.

(5) Once the drill described in US 2019/0321896 is mounted on the drill stand, then suitable dust management and lubrication management hoses may need to be attached to the drill. These hoses can interfere with the operation of the drill when installed on the drill stand. This may make setup, use and removal of the drill on the drill stand awkward and time consuming.

(6) The drill stand may be mounted on a workpiece surface. It is known to use a vacuum pump with the drill stand to provide a suction force against the workpiece surface as shown US2016/151934. A problem with this is that a control unit detects signals from an accelerometer to detect vibrations in the drill stand to infer a poor seal against the workpiece surface. However, if the drill experiences vibrations due to other issues, the control unit can incorrectly control the vacuum pump.

SUMMARY

(7) In a first aspect of the disclosure, there is provided a power tool stand for a removeable power tool having a first battery interface and a removeable battery having a first tool interface, the power tool stand comprising: a base; a projecting frame connected to the base; and a power tool carriage moveably mounted on the projecting frame, the power tool carriage comprising a second battery

interface and a second tool interface; wherein the second battery interface is compatible with the first tool interface and the second battery interface is arranged to electrically and mechanically connect together with first tool interface; and the second tool interface is compatible with the first battery interface and the second tool interface is arranged to electrically and mechanically connect together with first battery interface; and the first battery interface and the first tool interface are incompatible.

(8) Optionally, the power tool carriage comprises a battery adapter comprising the second battery interface.

(9) Optionally, the battery adapter is mountable to the power tool carriage.

(10) Optionally, the battery adapter comprises a carriage interface and the power tool carriage comprises an adaptor interface and adaptor interface is compatible with the carriage interface and the adaptor interface is arranged to electrically and mechanically connect together with carriage interface.

(11) Optionally, the carriage interface is the same as the second tool interface and the adaptor interface is the same as the first battery interface.

(12) Optionally, the battery adaptor is removably mountable to the power tool carriage.

(13) Optionally, the second battery interface is configured to mechanically and electrically connect to a battery having a voltage of at least 60V.

(14) Optionally, the power tool stand comprises a rack and pinion mechanism arranged to move the power tool carriage relative to the projecting frame.

(15) Optionally, the power tool carriage comprises a handle coupled to the rack and pinion mechanism.

(16) Optionally, the second battery interface is mounted on a first side of the power tool carriage and second tool interface is mounted on a second side of the power tool carriage.

(17) Optionally, the second tool interface is arranged project over the base.

(18) Optionally, the power tool stand comprises a plurality of the second battery interfaces arranged to electrically connect with a plurality of removeable batteries.

(19) Optionally, the power tool carriage comprises a latch recess for engaging with a latch mechanism mounted on the removeable battery.

(20) Optionally, the power tool carriage comprises an air duct connectable to a dust extractor.

(21) Optionally, the power tool carriage comprises a water duct connectable to a water supply.

(22) Optionally, the second battery interface is electrically connected to the second tool interface within the housing of the power tool carriage.

(23) In a second aspect of the disclosure there is provided a power tool stand system comprising: a power tool stand according to the first aspect; a removeable power tool having a first battery interface and the first battery interface is connectable to the second tool interface of the power tool stand; and a first removable battery having a first tool interface and the first tool interface is connectable to the second battery interface of the power tool stand.

(24) Optionally, the power tool stand system comprises a second battery having a third tool interface compatible with the first battery interface.

(25) In a third aspect of the disclosure there is provided a power tool stand comprising: a base; a projecting frame connected to the base; and a power tool carriage moveably mounted on the projecting frame; wherein the power tool carriage comprises a tool interface arranged to electrically connect with a removeable power tool and a first battery interface and a second battery interface, the first battery interface and the second battery interface being in electrical connection with the tool interface; wherein first battery interface and the second battery interface are respectively arranged to electrically connect with a first removeable battery and a second removeable battery.

(26) Optionally, the first battery interface and the second battery interface are configured to connect the first removeable battery and the second removeable battery in series.

(27) Optionally, the first battery interface and the second battery interface are configured to connect

the first removeable battery and the second removeable battery in parallel.

(28) Optionally, the power tool stand comprises a controller configured to selectively connect the first removeable battery and the second removeable battery to the power tool when the removable power tool, the first removeable battery, and the second removeable battery are mounted on the power tool carriage.

(29) Optionally, the controller is configured to connect the first removeable battery and/or the second removeable battery to the removable power tool in dependence of a received signal.

(30) Optionally, the received signal is a manually actuated signal indicating a user battery selection.

(31) Optionally, the received signal comprises a voltage indication of the first removeable battery and/or the second removeable battery and the controller is configured to connect the removeable power tool to the first removeable battery and/or the second removeable battery in dependence of the voltage indication of the first removeable battery and/or the second removeable battery.

(32) Optionally, the controller is configured to connect the first removeable battery and the second removeable battery to the removable power tool in series, in parallel or separately.

(33) Optionally, the controller is configured to compare the voltage indication of the first removeable battery and the second removeable battery and connect the first removeable battery or the second removeable battery in dependence of the voltage indication of the first removeable battery or the second removeable battery being closest to a required voltage of the removable power tool.

(34) Optionally, wherein the required voltage of the removable power tool is in dependence of the motor speed, a motor voltage rating, or a motor parameter.

(35) Optionally, the received signal comprises an indication that a status of either the first removeable battery or the second removeable battery has exceeded a threshold and the controller is configured to connect the motor to the other of the at least one first removeable battery or the at least one second removeable battery which has not exceeded the threshold.

(36) Optionally, the threshold is a battery operating temperature threshold, a battery capacity threshold, a battery voltage threshold, or a battery current threshold.

(37) Optionally, the received signal is signal indicating an error status of either the first removeable battery or the second removeable battery is malfunctioning or disconnected and the controller is configured to connect the removable power tool to the other of the first removeable battery or the second removeable battery.

(38) Optionally, the first battery interface and the second battery interface are mounted on a first side of the power tool carriage and tool interface is mounted on a second side of the power tool carriage.

(39) Optionally, the tool interface is arranged to project over the base.

(40) In a fourth aspect of the disclosure, there is provided a power tool stand comprising: a base; a projecting frame connected to the base; a power tool carriage moveably mounted on the projecting frame; a tool interface mounted on the power tool carriage arranged to couple with a removeable power tool; an electrical interface mounted on the power tool carriage arranged to electrically couple with a removeable battery; wherein the power tool stand comprises at least one duct configured to convey a fluid; and at least one pump in fluid communication with the fluid duct wherein the at least one pump is in electrical connection with the electrical interface.

(41) Optionally, the at least one duct is connected to a fluid connector configured to engage with a power tool fluid duct when the removeable power tool is mounted to the power tool carriage.

(42) Optionally, the at least one duct is in fluid communication with a water supply.

(43) Optionally, the water supply is a non-pressurised water supply.

(44) Optionally, the water supply is a water reservoir.

(45) Optionally, power tool carriage comprises a switch in electrical connection with the at least one pump configured to selectively control actuation of the at last one pump.

(46) Optionally, the switch is configured to selectively control actuation of the at last one pump

when the removeable power tool and the removeable battery are mounted on the power tool carriage.

(47) Optionally, the at least one pump is mounted in the power tool carriage.

(48) Optionally, the at least one duct is mounted on the power tool carriage.

(49) Optionally, the power tool stand comprises a controller configured to selectively control actuation of the at last one pump.

(50) Optionally, the tool interface comprises a power tool electrical interface mounted on the power tool carriage arranged to electrically couple with the removeable power tool.

(51) Optionally, the power tool electrical interface is in electrical connection with the electrical interface.

(52) In a fifth aspect of the disclosure, there is provided a power tool stand comprising: a base mountable on a workpiece surface; a frame connected to the base; a power tool carriage mounted on the frame; a tool interface mounted on the power tool carriage arranged to secure a removeable power tool; a work surface seal mounted on the base and configured to engage the workpiece surface and in fluid communication with a vacuum source and the work surface seal is configured to generate a vacuum seal against the workpiece surface when the vacuum source is actuated; a pressure sensor in fluid communication with the work surface seal and configured to generate a pressure signal; and a controller configured to determine when the vacuum pressure in work surface seal is decreasing in dependence of the received pressure signal from the pressure sensor and issue one or more control signals based on a detected decreased vacuum pressure in the work surface seal; wherein the power tool carriage comprises a battery interface arranged to secure a removeable battery and the battery interface is electrically connected to the tool interface.

(53) Optionally, the controller is configured to send a control signal to the removeable power tool.

(54) Optionally, the controller is configured to issue a first control signal in dependence of determining that the vacuum pressure in the work surface seal falls below a first vacuum pressure threshold.

(55) Optionally, the first control signal is configured to issue an alert to the user when the vacuum pressure in the work surface seal falls below a first vacuum pressure threshold.

(56) Optionally, the controller is configured to issue a second control signal in dependence of determining that the vacuum pressure of the work surface seal falls below a second vacuum pressure threshold, wherein the second vacuum pressure threshold is lower than the first vacuum pressure threshold.

(57) Optionally, the second control signal is configured deactivate the removable power tool.

(58) Optionally, the power tool carriage comprises a switch in electrical connection with the tool interface and the battery interface configured to selectively control actuation of the removeable power tool and the controller is configured to actuate the switch.

(59) Optionally, the vacuum source is a vacuum pump.

(60) Optionally, the vacuum pump is mounted in the power tool carriage or on the base.

(61) Optionally, the vacuum pump is electrically connected to the removeable battery when the removeable battery is mounted to the battery interface.

(62) Optionally, the controller is configured to issue a control signal to the vacuum pump to increase the vacuum pressure in the work surface seal.

(63) Optionally, the pressure sensor is wirelessly connected or in wired connection with the controller.

(64) Optionally, the work surface seal is configured mount to a vertical surface.

(65) In a sixth aspect of the disclosure, there is provided a power tool stand comprising: a base mountable on a workpiece surface; a frame connected to the base; a power tool carriage mounted on the frame; a tool interface mounted on the power tool carriage arranged to secure a removeable power tool; a work surface seal mounted on the base and configured to engage the workpiece surface and in fluid communication with a vacuum source and the work surface seal is configured

to generate a vacuum seal against the workpiece surface when the vacuum source is actuated; wherein the vacuum source is a vacuum pump mounted in the power tool stand.
(66) Optionally, the vacuum pump is mounted in the power tool carriage.
(67) Optionally, the vacuum pump is mounted in the base.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Various other aspects and further examples are also described in the following detailed description and in the attached claims with reference to the accompanying drawings, in which:
- (2) FIG. 1 shows a perspective view of a power tool stand according to an example;
- (3) FIG. 2 shows a perspective view of part of a power tool stand according to an example;
- (4) FIG. 3 shows a side view of an exemplary power tool mountable on a power tool stand according to an example;
- (5) FIG. 4 shows a side view of a power tool stand according to an example;
- (6) FIG. 5 shows a side view of part of a power tool stand according to an example;
- (7) FIGS. 6a and 6b respectively show a partial front and rear view of a power tool stand according to an example;
- (8) FIG. 7 shows a side view of part of a power tool stand according to an example;
- (9) FIG. 8 shows a plan view of part of a power tool stand according to an example;
- (10) FIG. 9 shows a side view of a power tool stand according to an example;
- (11) FIG. 10 shows a side view of a power tool stand according to an example;
- (12) FIG. 11 shows a schematic view of a power tool stand according to an example;
- (13) FIG. 12 shows a graph of pressure of a workpiece surface seal over time according to an example;
- (14) FIG. 13 shows a flow diagram of operation of a power tool stand according to an example;
- (15) FIG. 14 shows a schematic view of a power tool stand according to an example; and
- (16) FIG. 15 shows a flow diagram of operation of a power tool stand according to an example.

DETAILED DESCRIPTION

- (17) FIG. 1 shows a perspective view of a power tool stand **100** according to an example. The power tool stand **100** is arranged to receive a removeable power tool **102**. The removeable power tool **102** can be mounted and secured in the power tool stand **100** as required by the user. The term “removeable” refers to the function of the removeable power tool **102** being selectively removeable from the power tool stand **100**. In other words, the removeable power tool **102** is not permanently fixed to the power tool stand **100**.
- (18) The power tool stand **100** comprises a base **104** and a projecting frame **106** fixed to the base **104**. A power tool carriage **108** is moveably mounted on the projecting frame **106** and moveable along a longitudinal axis A-A of the projecting frame **106**. By moving the power tool carriage **108** along the projecting frame **106**, the distance between the power tool carriage **108** and the base **104** can be varied. The projecting frame **106** projects through a through hole **202** (best shown in FIG. 2) in the power tool carriage **108**. FIG. 2 shows a perspective view of the power tool carriage **108** of the power tool stand **100**.
- (19) The power tool carriage **108** is optionally moveably coupled to projecting frame **106** via a carriage mechanism (not shown). In some less preferred examples (not shown in the Figs), the power tool carriage **108** is fixed with respect to the projecting frame **106** and does not move with respect to the power tool carriage **108**.
- (20) The projecting frame **106** as shown in FIG. 1 is substantially upright. In some examples, the power tool stand **100** can be used in the orientation as shown in FIG. 1 and in this case, the projecting frame **106** will remain substantially upright positioned on a workpiece surface **920** of a

workpiece **918**. The workpiece surface **920** is best shown in FIG. **8**. FIG. **8** shows a plan view of the power tool carriage **108** of the power tool stand **100**.

(21) However, in some other examples, the power tool stand **100** can be used in a substantially horizontal orientation. In this case, the projecting frame **106** will be substantially horizontal. Indeed, the power tool stand **100** may be used in any orientation as required and the position of the projecting frame **106** will be adjusted accordingly.

(22) The projecting frame **106** as shown in FIG. **1** is a single post projecting up from the base **104**. However, in other examples the projecting frame **106** can optionally comprise a plurality of posts (not shown) projecting up from the base **104** and the power tool carriage **108** may be mounted on the plurality of posts. The projecting frame **106** as shown in FIG. **1** is mounted perpendicular to the plane of the base **104** at a coupling **110**. The plane of the base **104** can be substantially parallel with the plane of a workpiece **918** (best shown in FIG. **9**). FIG. **9** shows a side view of the power tool stand **100** with the removeable power tool **102** mounted thereon.

(23) In some examples, the coupling **110** optionally comprises a pivot connection and the projecting frame **106** is pivotable with respect to the base **104**. In this way, the projecting frame **106** can be angled with respect to a normal axis (not shown) of the plane of a workpiece **918**. The normal axis of the plane of the workpiece **918** and the base **104** is parallel with the longitudinal axis A-A as shown in FIG. **1**.

(24) The coupling **110** comprises a screw fastener **112** threaded through arcuate slots **114** in a pair of projecting plates **116** fixed to the base **104**. The screw fastener **112** is screwed into the projecting frame **106** or into a reciprocal fastening nut **118** (as shown in FIG. **2**) on the opposite side of the projecting frame **106**. When the screw fastener **112** and reciprocal fastening nut **118** are tightened against the projecting plates **116** and the projecting frame **106**, frictional forces keep the projecting frame **106** fixed with respect to the base **104**. Other fastening mechanisms e.g. clips, clamps, or any other suitable means can be used to fix the projecting frame **106** with respect to the base **104**.

(25) The pair of projecting plates **116** each comprises an arcuate slot **114** and form a yoke for receiving a portion of the projecting frame **106** between the projecting plates **116**. The yoke as shown in FIG. **1** is optional and other pivot connections can be provided. For example a single projecting plate **116** can be used to couple the base **104** to the projecting frame **106**. In some examples, the arcuate slot **114** optionally comprises a plurality of indexing positions (not shown) for positioning the reciprocal fastening nut **118** and the screw fastener **112** at predetermined positions along the arcuate slot **114**.

(26) Optionally, the projecting plates **116** comprise an inclination scale (not shown) to indicate the angle of the projecting frame **106** with respect to the plane of the base **104**. In some examples the projecting frame **106** is arranged to pivot between 0 to 60 degrees from a normal axis (not shown) perpendicular to the base **104**. In other examples, the projecting frame **106** can pivot any angle with respect to a normal axis perpendicular to the base **104**.

(27) In some examples, the projecting frame **106** is alternatively permanently fixed with respect to the base **104**. In this case the projecting frame **106** cannot be adjusted with respect to the base **104**. For example, the projecting frame **106** is welded to the base **104** in the upright position as shown in FIG. **1**.

(28) In some examples, the projecting frame **106** can be pivotable with respect to base **104** such that the longitudinal axis A-A of the projecting frame **106** can be positioned parallel with the plane of the base **104**. This can be useful for transporting the power tool stand **100** to the work site.

(29) In some examples, the projecting frame **106** comprises a reinforced cross-sectional shape for increasing rigidity of the projecting frame **106**. The projecting frame **106** can optionally comprise an I-beam cross-sectional shape in some examples.

(30) In some examples, the base **104** comprise one or more fixing bolts **120** for anchoring the base **104** of the power tool stand **100** to the workpiece **918**. FIG. **1** shows the base **104** comprising four fixing bolts **120** located at each corner of the base **104**. In some examples, there can be additional

or fewer fixing bolts **120** for anchoring the base **104** to the workpiece **918** as required.

(31) The fixing bolts **120** allow the power tool stand **100** to be securely fastened to a workpiece **918** surface in a horizontal plane, a vertical plane, or an inclined surface. For example a user may using the fixing bolts **120** to anchor the power tool stand **100** to a floor or a wall before operation.

(32) When the user anchors the base **104** to a vertical surface such as a wall (not shown), the user may disassemble the power tool stand **100** and remove the projecting frame **106** from the base **104**. Once the user has anchored the base **104** to the wall, the user may then reattach the projecting frame **106** back on the base **104**.

(33) The base **104** optionally comprises wheels **122** for transporting the power tool stand **100** to the worksite. The wheels **122** are mounted on the base **104** such that the wheels **122** do not project beyond an engaging surface **134** of the base **104**. The engaging surface **134** is arranged to abut the workpiece surface **920**. Accordingly, the wheels **122** do not touch the workpiece surface **920** when the engaging surface **134** abuts the workpiece surface **920**. In some examples, the wheels **122** are optionally removeable. This means the user can remove the wheels **122** when placing the power tool stand **100** in tight spaces on a wall or a floor.

(34) The base **104** can optionally further comprise one or more stand handles (not shown) for also assisting transportation and handling. The user can optionally grasp the stand handle mounted on the base **104** and the projecting frame **106** in order to move the power tool stand **100** into position.

(35) The power tool carriage **108** and power tool stand **100** will be discussed in more detail with respect to FIGS. **1** and **2**. As shown in FIG. **2**, the power tool carriage **108** is arranged to receive a removeable power tool **102** and a removeable battery **130**. The projecting frame **106** allows a power tool carriage **108** to be precisely moved along the longitudinal axis A-A and then fixed with respect to the projecting frame **106**.

(36) The power tool carriage **108** comprises a carriage housing **200**. The carriage housing **200** comprises a clam shell type construction having two halves which are fastened together. The halves of the carriage housing **200** are fastened together with screws but in alternative examples any suitable means for fastening the carriage housing **200** together may be used such as glue, clips, bolts and so on. For the purposes of clarity, the fastenings in the carriage housing **200** are not shown in FIG. **2**. The carriage housing **200** can comprise a unitary element surrounding the internal components of the power tool carriage **108**. In other examples, the carriage housing **200** can comprise one or more housing portions (not shown) which are mounted together to form the carriage housing **200**.

(37) The power tool carriage **108** comprises a power tool interface **204** arranged to electrically and mechanically connect the removeable power tool **102** to the power tool carriage **108**.

(38) The power tool carriage **108** further comprises a battery interface **206** arranged to electrically and mechanically connect the removeable battery **130** to the power tool carriage **108**.

(39) The power tool interface **204** and the battery interface **206** are electrically connected together via one or more internal wires (not shown) routed inside the carriage housing **200**. The internal wires can be for example a flexible connection ribbon arranged to electrically couple pairs of electrical contacts on the power tool interface **204** and the battery interface **206**. In other examples, power tool interface **204** and the battery interface **206** can be electrically connected by separate wires, a printed circuit board or any other suitable means for electrically connecting the power tool interface **204** and the battery interface **206**.

(40) In this way, the removeable power tool **102** and the removeable battery **130** are respectively electrically connected to the power tool interface **204** and the battery interface **206** when mounted to the power tool carriage **108**. This means that when mounted to the power tool carriage **108**, the removeable battery **130** can supply power to the removeable power tool **102**. Accordingly, the removeable power tool **102** can be powered by the removeable battery **130** when both mounted on the power tool carriage **108** as if the removeable battery **130** and the removeable power tool **102** are directly coupled together.

(41) In some examples operation of the removeable power tool **102** when mounted on the power tool stand **100** is no different from when the removeable power tool **102** is directly connected to the removeable battery **130** as shown in FIG. 3. FIG. 3 shows a side view of the removable power tool **102** with a removeable battery **130** mounted to the removeable power tool **102**. The user can therefore intuitively use the removeable power tool **102** when mounted on the power tool carriage **108**.

(42) Turning to FIG. 3, the removeable power tool **102** will now be discussed in more detail. FIG. 3 shows a side view of an the removeable power tool **102**.

(43) The removeable power tool **102** as shown in FIG. 3 is an example of a removeable power tool **102** that is mountable on the power tool stand **100**. The removeable power tool **102** as shown in FIG. 3 in some examples is a drill. In particular, the removeable power tool **102** is a diamond core drill. In other examples, the removeable power tool **102** can be hammer drill, a router, a reciprocating saw, plunge saw, a circular saw, an impact driver, a drill, a hammer drill, a multitool, an oscillating tool, a rotary hammer, a chipping hammer, or any other similar power tool mountable on a power tool stand **100**.

(44) The removeable power tool **102** is battery powered and comprises a tool mounted battery interface **300** for engaging with the reciprocal battery mounted tool interface **310** on the removeable battery **130**. The removeable power tool **102** is generally elongate and comprises a longitudinal axis B-B. The removeable power tool **102** and the removeable battery **130** are engageable along a third interface plane C-C. The third interface plane C-C is parallel with the longitudinal axis B-B of the removeable power tool **102**. This means that the removeable battery **130** is mounted on the tool mounted battery interface **300** by sliding the removeable battery **130** in a direction D indicated by the arrow in FIG. 3. The direction D is away from the nose **302** of the removeable power tool **102** and towards a rear power tool handle **304** located at the rear **306** of the removable power tool **102**.

(45) In order to mount the removeable power tool **102** and the removeable battery **130** on the power tool stand **100**, the removeable battery **130** is disengaged from the removeable power tool **102**. This is achieved by reversing the sliding action described above in reference to FIG. 3. The separate removeable power tool **102** and the removeable battery **130** can then be individually mounted on the power tool carriage **108**.

(46) As discussed above, both the removeable power tool **102** and the removeable battery **130**, are mechanically secured to the power tool carriage **108**. This means that the power tool carriage **108** moves in unison with the removeable power tool **102** and the removeable battery **130**. This can help the user stabilise the removeable power tool **102** during use and improves the quality of the cut in the workpiece **918**.

(47) The power tool interface **204** mounted on the power tool carriage **108** is compatible with the tool mounted battery interface **300**. Likewise the battery interface **206** on the power tool carriage **108** is compatible with the battery mounted tool interface **310**.

(48) However, in some examples, the tool mounted battery interface **300** on the removeable power tool **102** is not compatible with the battery mounted tool interface **500** on the removeable battery **502** (e.g. as shown in FIG. 5).

(49) For example, the removeable battery **502** as shown in FIG. 5 is a high capacity removeable battery **502** configured to power the removeable power tool **102** when mounted on the power tool stand **100**. The high capacity removeable battery **502** may not be suitable for connecting directly to the removeable power tool **102** because the high capacity removeable battery **502** is too large and heavy. In some examples, the mechanical and electrical interfaces of the battery mounted tool interface **500** on the high capacity removeable battery **502** are different from the mechanical and electrical interfaces on the tool mounted battery interface **300** on the removeable power tool **102**. That is, the removeable battery **103** as shown in FIG. 3 is different from the high capacity removeable battery **502** as shown in FIG. 5.

(50) This means that the high capacity removeable battery **502** as shown in FIG. 5 comprises an electrical and mechanical lock out preventing mounting of the high capacity removeable battery **502** on the removable power tool **102**. This means only the correct form, shape, and specification of the removeable battery **130** and the high capacity removeable battery **502** can be mounted to the removeable power tool **102** and the power tool stand **100** respectively.

(51) For example, the removeable battery **130** as shown in FIG. 1 is mountable directly to the removeable power tool **102**, but the high capacity removeable battery **502** as shown in FIG. 5 is not mountable directly to the removable power tool **102**. Instead, the high capacity removeable battery **502** as shown in FIG. 5 is mountable directly to power tool carriage **108**.

(52) In some examples, the power tool carriage **108** optionally comprises a battery adaptor **504**. The battery adaptor **504** is configured to mechanically and electrically connect to the high capacity removeable battery **502**. The battery adaptor **504** is configured mechanically and electrically connect to the power tool carriage **108**. When the battery adaptor **504** is mounted to the power tool carriage **108**, the battery adaptor **504** is electrically connected to the power tool interface **204**. In some other examples, the battery adaptor **504** is integral with the power tool carriage **108**.

(53) The battery adaptor **504** comprises an adaptor battery interface **702** configured to electrically and mechanically couple with the battery mounted tool interface **500** on the high capacity removeable battery **502**.

(54) The battery adaptor **504** in some examples is of a modular construction and is removably mountable to the power tool carriage **108**. The battery adaptor **504** is mountable to the power tool carriage **108** and is fixed in place with one or more screw fastenings **700**. This means that the battery adaptor **504** is fixed to the power tool carriage **108** during normal use. However, during manufacture or maintenance, the battery adaptor **504** can be replaced with another different battery adaptor **504**. For example, the battery adaptor **504** as shown in FIG. 5 is suitable for e.g. a 60V high capacity removeable battery **502**. In other examples, the battery adaptor **504** is configured to mechanically and electrically connect to any shape, size and capacity of battery. Another battery adaptor (not shown) can be replaced for the battery adaptor **504** as shown in FIG. 5 to connect a different battery to the power tool carriage **108**. The different battery can have a different shape, size, voltage from the high capacity removeable battery **502** as shown in FIG. 5 or the removeable battery **130** as shown in FIG. 1.

(55) The removeable battery adaptor **504** as shown in FIGS. 5 and 7 comprises a carriage interface (not shown) and the power tool carriage **108** comprises an adaptor interface (not shown) and carriage interface is compatible with the adaptor interface. This means that the carriage interface on the battery adaptor **504** and the adaptor interface on the power tool carriage **108** are arranged to electrically and mechanically connect together. This means that the battery adaptor **504** can be conveniently plugged into the power tool carriage **108** and makes manufacture easier.

(56) In some examples, the carriage interface on the battery adaptor **504** is the same as the power tool interface **204** on the power tool carriage **108**. Furthermore in some examples the adaptor interface on the power tool carriage **108** is the same as the tool mounted battery interface **300** on the removeable power tool **102**. In other words, when the battery adapter **504** is removed from the power tool carriage **108**, the removeable battery **130** as shown in FIG. 1 which is mountable on and compatible with the removable power tool **102** is mountable on the power tool carriage **108**.

(57) When the tool mounted battery interface **300** on the removeable power tool **102** is not compatible with the battery mounted tool interface **500**, this means that the different methods of attaching the high capacity removeable battery **502** to power tool carriage **108** must be used compared to the method of attaching the removeable power tool **102** to the power tool carriage **108**.

(58) In some examples, the battery adaptor **504** comprises a latch mechanism **506**. The latch mechanism **506** comprises a spring biased hook portion (not shown) configured to engage a reciprocal recess in the high capacity removeable battery **502**. The latch mechanism **506** secures the high capacity removeable battery **502** to the battery adaptor **504**. The latch mechanism **506** is

depressed to release the spring biased hook portion from the recess and the high capacity removeable battery **502** is then removeable from the battery adaptor **504**. Alternatively, the latch mechanism **506** can be mounted on the high capacity removeable battery **502** and the reciprocal recess is located on the battery adaptor **504**.

(59) Mounting the removable power tool **102** and the removeable battery **130** to the power tool stand **100** will be discussed in more detail with reference to FIGS. **4**, **6a** and **6b**. FIG. **4** shows the longitudinal axis A-A of the projecting frame **106**. Furthermore, FIG. **4** shows a first interface plane F-F aligned with a plane of the battery interface **206** mounted on the power tool carriage **108**. The removeable battery **130** is mountable on the power tool carriage **108** in a direction E represented by the arrow. In some examples, the removeable battery **130** is mountable in the direction E towards the base **104**. In some examples, the first interface plane F-F is substantially parallel with the longitudinal axis A-A of the projecting frame **106**.

(60) FIG. **4** also shows a second interface plane H-H aligned with a plane of the power tool interface **204** mounted on the power tool carriage **108**. The removeable power tool **102** is mountable on the power tool carriage **108** in a direction G represented by the arrow. In some examples, the removeable power tool **102** is mountable in a direction G towards the base **104**. In some examples, the second interface plane H-H is substantially parallel with the longitudinal axis A-A of the projecting frame **106**. In some examples, the first and second interface planes are substantially parallel.

(61) As shown in FIG. **4**, the removeable battery **130** and the removeable power tool **102** are optionally mountable in the same direction E, G with respect to the power tool carriage **108**. This is advantageous because the user can use a similar pushing action in the same direction E, G to mount both the removeable battery **130** and the removeable power tool **102** on the power tool stand **100**. For example, the user can stand over the power tool stand **100** as shown in FIG. **1** and push down both the removeable power tool **102** and the removeable battery **130** to mount them on the power tool carriage **108**. This can make assembly of the removeable power tool **102** and the removeable battery **130** on the power tool carriage **108** easier and more intuitive.

(62) In order to mount the removeable battery **130** and the removeable power tool **102** on the power tool carriage **108**, in some examples, both the removeable battery **130** and the removeable power tool **102** slide into mechanical engagement with the power tool carriage **108**. The sliding action of the removeable battery **130** and the removable power tool **102** continues until a latch mechanism (not shown in FIG. **4**) engages between the power tool carriage **108** and the removeable battery **130** and/or the removeable power tool **102**. In some examples, the battery interface **206** and the power tool interface **204** comprise two latch mechanisms which are the same as the latch mechanism used to hold the removeable battery **130** in engagement when the mounted to the removeable power tool **102**. The latch mechanism in some examples is the same as shown in reference to the latch mechanism **506** shown in FIG. **5** on the battery adaptor **504**.

(63) The direction E, as shown in FIG. **4**, is the direction that the removeable battery **130** is mounted on the power tool carriage **108**. The direction E is in a direction opposite to the direction D that the removeable battery **130** is mounted on the removeable power tool **102**, as shown in FIG. **3**.

(64) Mounting the removeable power tool **102** and the removeable battery **130** on the power tool stand **100** will now be discussed in more detail in reference to FIGS. **6a**, and **6b**. FIGS. **6a** and **6b** respectively show a partial front and rear view of a power tool stand **100**.

(65) FIGS. **6a** and **6b** respectively show the power tool interface **204** and the battery interface **206** without the removeable power tool **102** or the removeable battery **130** mounted thereon for the purposes of clarity.

(66) The power tool interface **204** as shown in FIG. **6a** comprises a power tool mechanical interface **600** and a power tool electrical interface **602**. In some examples, the power tool mechanical interface **600** and the power tool electrical interface **602** are integral with the carriage

housing **200**. For example, the power tool mechanical interface **600** and the power tool electrical interface **602** are moulded into the carriage housing **200**. In some examples, the power tool mechanical interface **600** and the power tool electrical interface **602** are positioned together or near each other on the carriage housing **200**. In some other less preferred examples, the power tool mechanical interface **600** and the power tool electrical interface **602** are located remote from each other on the carriage housing **200**.

(67) As shown in FIG. **6a**, the power tool mechanical interface **600** optionally comprises a pair of rails **604**, **606** configured to engage with reciprocal slots mounted on the removeable power tool **102**. The pair of rails **604**, **606** slide into engagement with the reciprocal slots and secure the removeable power tool **102** to the power tool carriage **108**.

(68) The power tool electrical interface **602** optionally comprises a plurality of electrical contact clips **608** configured to engage a plurality of reciprocal projecting contact ribs (not shown) mounted on the removeable power tool **102**. For the purposes of clarity only one electrical contact clip **608** is labelled. The power tool mechanical interface **600** and the power tool electrical interface **602** are one optional way of connecting the removeable power tool **102** to the power tool carriage **108**.

(69) The battery interface **206** as shown in FIG. **6b** comprises a battery mechanical interface **610** and a battery electrical interface **612**. In some examples, the battery mechanical interface **610** and the battery electrical interface **612** are integral with the carriage housing **200**. For example, the battery mechanical interface **610** and the battery electrical interface **612** are moulded into the carriage housing **200**. In some examples, the battery mechanical interface **610** and battery electrical interface **612** are positioned together or near each other on the carriage housing **200**. In some other less preferred examples, the battery mechanical interface **610** and the battery electrical interface **612** are located remote from each other on the carriage housing **200**.

(70) The battery mechanical interface **610** optionally comprises a pair of slots **614**, **616** configured to engage with reciprocal rails mounted on the removeable battery **130**. The battery electrical interface **612** comprises a plurality of projecting electrical contact ribs **618** configured to engage a plurality of reciprocal electrical contact clips (not shown) mounted on the removeable battery **130**. For the purposes of clarity only one projecting electrical contact rib **618** is labelled in FIG.

(71) **6b**. The battery mechanical interface **610** and the battery electrical interface **612** are one optional way of connecting the removeable power tool **102** to the power tool carriage **108**.

(72) In some examples, each of the power tool mechanical interface **600** and the battery mechanical interface **610** can optionally comprises a plurality of rails and slots to further improve the mechanical engagement when mounting the removeable power tool **102** and the removeable battery **130** on the power tool carriage **108**. For example, each of the power tool mechanical interface **600** and the battery mechanical interface **610** can each comprise both slots and rails arranged to respectively engage with reciprocal rails and slots.

(73) Alternatively, other mechanisms and features can be used for securely connecting the removeable power tool **102** to the power tool carriage **108**. For example, in some less preferred alternatives, the rail and slot arrangement as discussed above is not used for mechanically securing the removeable battery **130** and the removeable power tool **102** to the power tool carriage **108**. In this case, other external features on the removeable battery **130** and the removeable power tool **102** are used to securely fasten the removeable battery **130** and the removeable power tool **102** to the power tool carriage **108**. In some examples, latches, straps, clips, clamps or any other suitable securing mechanism can be used to secure the removeable battery **130** and the removeable power tool **102**.

(74) As mentioned above, in some less preferred examples, the battery interface **206** comprises a battery mechanical interface **610** and a battery electrical interface **612** in separate locations on the power tool carriage **108**. Similarly, the power tool interface **204** comprises a power tool mechanical interface **600** and a power tool electrical interface **602** in separate locations on the power tool

carriage **108**. This can be optionally achieved by making a separate electrical connection not using the previously discussed electrical contacts. For example, the battery electrical interface **612** and the power tool electrical interface **602** can be flexible ribbons projecting through the carriage housing **200** and the flexible ribbons have a plurality of electrical contacts connected thereto. The flexible ribbons can be manually attached to the ensure an electrical connection with the removeable power tool **102** and the removeable battery **130**.

(75) In another less preferred example, the removeable battery **130** can be mounted on the power tool carriage **108** in a different direction e.g. opposite or inclined to the direction that the removeable power tool **102** is mounted on the power tool carriage **108**.

(76) The removeable power tool **102** is optionally mechanically connected to the power tool carriage **108** only at the power tool mechanical interface **600**. However, in some other examples as shown in e.g. FIG. 2, the removeable power tool **102** is mounted to the power tool carriage **108** at a plurality of other mechanical couplings in addition to the power tool mechanical interface **600**.

(77) FIGS. 4, **6a** and **6b** discuss mounting the removeable battery **130** to the power tool carriage **108** and the removeable power tool **102** to the power tool carriage **108**. However similar electrical and mechanical features discussed in reference to FIGS. 4, **6a** and **6b** can also be used to mount the high capacity removeable battery **502** to the battery adaptor **504**. In addition similar electrical and mechanical features discussed in reference to FIGS. 4, **6a** and **6b** can be used to mount the battery adaptor **504** to the power tool carriage **108** in the example when the battery adaptor **504** is removably mountable to the power tool carriage **108**.

(78) In some other examples as shown in e.g. FIG. 1 and FIG. 2, the removeable power tool **102** is mounted to the power tool carriage **108** at a plurality of other mechanical couplings in addition to the power tool mechanical interface **600**. Another mechanical coupling will now be discussed in reference to FIG. 2.

(79) The power tool carriage **108** optionally comprises a tool clamp **214** configured to secure the removeable power tool **102** to the power tool stand **100**. The tool clamp **214** is configured to engage a nose portion **126** (best shown in FIG. 1) of a housing **124** of the removeable power tool **102**. The tool clamp **214** may optionally be secured to the nose portion **126** of the removeable power tool **102**. When the tool clamp **214** is engaged with the nose portion **126**, the removeable power tool **102** is further mechanically coupled to the power tool carriage **108**. The coupling between the removeable power tool **102** and the power tool carriage **108** will be discussed in more detail below.

(80) The tool clamp **214** optionally comprises a first clamp arm **216** and a second clamp arm **218** configured to urge towards each other. The first clamp arm **216** and the second clamp arm **218** each comprise a clamp jaw **220**. The clamp jaws **220** on the first and second clamp arms **216**, **218** are configured to engage a reciprocal housing recess in the nose portion **126** of the housing **124** of the removeable power tool **102**. When the first and second clamp arms **216**, **218** are engaged with the nose portion **126** between the first and second clamp arms **216**, **218**, the first and second clamp arms **216**, **218** exert a gripping force against the nose portion **126**.

(81) By mounting the removeable power tool **102** to the power tool carriage **108** with the tool clamp **214**, the removable power tool **102** does not move with respect to the power tool carriage **108**. This means that the user can have the confidence that the removeable power tool **102** is securely mounted to the power tool stand **100** when performing a cutting operation.

(82) As can be seen from FIG. 4, the removeable battery **130** is mounted on a first side **400** of the power tool carriage **108**. The removeable power tool **102** is mounted on a second side **402** of the power tool carriage **108**. The first side **400** and the second side **402** of the power tool carriage **108** are positioned either side of the projecting frame **106**. Accordingly the first side **400** the second side **402** of the power tool carriage **108** are on opposite sides of the power tool carriage **108**.

(83) The power tool carriage **108** conveniently positions the removeable battery **130** remote from the removeable power tool **102** when both mounted on the power tool carriage **108**. This means

that the weight of the removeable power tool **102** and the removeable battery **130** can be better distributed about the projecting frame **106**. For example, the removeable power tool **102** can be mounted closer to the projecting frame **106**. This reduces the turning moment of the removeable power tool **102** when mounted on the power tool stand **100**. This makes the power tool stand **100** more stable when the removeable power tool **102** is mounted on the power tool carriage **108**.

(84) By mounting the removeable battery **130** on an opposite side of the power tool carriage **108** on a different side of the projecting frame **106**, the weight of the removeable battery **130** provides a counterweight to the weight of the removeable power tool **102**. This further improves the stability of the power tool stand **100** when the removable power tool **102** and the removeable battery **130** are mounted on the power tool stand **100**.

(85) The removeable battery **130** as shown in the accompanying Figs is only a single battery mounted on the power tool carriage **108**. However, in other examples, the power tool carriage **108** comprises a plurality of battery interfaces **204** mounted on the power tool carriage **108** configured to electrically and mechanically connected a plurality of removeable batteries **130**. This will be discussed in more detail below with respect to FIG. **8**.

(86) The carriage mechanism (not shown) is mounted within the carriage housing **200** and is arranged to move the power tool carriage **108** with respect to the projecting frame **106** when the carriage mechanism is actuated. The projecting frame **106** projects through a through hole **202** in the carriage housing **200**.

(87) In some examples, the carriage mechanism comprises a rack (not shown) and pinion mechanism (not shown) for moving the power tool carriage **108** with respect to the projecting frame **106**. A rack is mounted on one side of the projecting frame **106** along the length of the projecting frame **106**. A drive gear is mounted on a drive shaft and a mechanism handle **128** is mounted on the drive shaft for manually rotating the drive shaft. The drive gear is optionally rotatably coupled to an intermediate gear and the intermediate gear engages the rack. In some other examples, the drive gear directly engages with the rack. In some other examples, any other suitable mechanism can be used to move the power tool carriage **108** with respect to the projecting frame **106**. In some other examples, the mechanism handle **128** is alternatively or additionally replaced with a drive mechanism coupled to a motor (not shown) for assisting the movement of the power tool carriage **108** on the power tool stand **100**.

(88) Turning to FIG. **9**, another example will now be described. FIG. **9** shows a side view of the power tool stand **100**. In some examples, the power tool stand **100** optionally comprises at least one accessory electrical interface **900** configured to connect to at least one accessory. The at least one accessory electrical interface **900** in some examples is an electrical socket **900**. The electrical socket **900** is shown in FIG. **9** as a standard mains power socket, however, the electrical socket **900** in some examples is a low voltage or a DC voltage socket. In some examples, the electrical socket **900** is adapted for non-mains power plugs so that only low voltage or DC voltage accessories can be inserted into the electrical socket **900**.

(89) The electrical socket **900** is electrically connected to the battery interface **206** and/or the power tool interface **204**. By connecting the electrical socket **900** to the battery interface **206**, the removeable battery **130** can provide power to the electrical socket **900** when the removeable battery **130** is mounted to the battery interface **206**.

(90) In some examples, the accessory that is connected to the electrical socket **900** is a water pump **1000** (best shown in FIG. **10**) and/or a vacuum source **904** e.g. a vacuum pump **904** (best shown in FIG. **9**). In other examples, the accessory can be any other suitable accessory such as a lamp, worksite radio or any other equipment used in the worksite.

(91) The electrical socket **900** is optional and in some other examples is not provided. In this case, one or more accessories e.g. water pump **1000** (best shown in FIG. **10**) and/or a vacuum source **904** e.g. a vacuum pump **904** (best shown in FIG. **9**) are optionally powered from the removeable battery **130** via a power connection **922**. FIG. **9** schematically shows a power connection **922**

between the removable battery **130** and the vacuum pump **904**. Similarly the water pump **1000** as shown in FIG. **10** can also be powered from the removable battery **130** via the power connection **922** (but the power connection **922** is not shown in FIG. **10** for the purposes of clarity). Whilst FIG. **9** and FIG. **10** show different examples of the power tool stand **100** with different accessories (e.g. the vacuum pump **904** and the water pump **1000**) in another example, the power tool stand **1000** can comprise both the vacuum pump **904** and the water pump **1000**. In this case, both the vacuum pump **904** and the water pump **1000** can be powered from the removable battery **130** (or the high capacity removable battery **502**).

(92) Operation of the water pump **1000** and the vacuum pump **904** will be discussed below in more detail in reference to FIGS. **9** and **10** respectively.

(93) In an alternative example, the accessory electrical interface **900** is optionally connectable to an external power source and the accessory electrical interface **900** is electrically connected to the first electrical interface and/or the second electrical interface. In some examples, the external power source is a mains power supply or an external battery. When a mains power supply is connected to the accessory electrical interface **900**, the mains power supply is arranged to power the removable power tool **102** and/or charge the removable battery **130** when the removable power tool **102** and the removable battery **130** are mounted on the power tool carriage **108**.

(94) In some examples the switch **902** is an ON/OFF switch for the removable power tool **102**. In some examples, the switch **902** is in series with an ON/OFF switch e.g. the power trigger switch **308** on the removable power tool **102**. In this way, the switch **902** must be actuated before the removable power tool **102** is actuated. This reduces the chances of the removable power tool **102** being accidentally actuated when mounted on the power tool carriage **108**.

(95) In some other examples, the switch **902** is alternatively an actuation switch configured to actuate the water pump **1000** independently of the removable power tool **102**.

(96) In some examples, the switch **902** is configured to wirelessly actuate the water pump **1000** or a vacuum pump **904**. In some examples, the switch **902** is configured to control a transmitter-receiver (not shown) mounted in the power tool carriage **108**. The transmitter-receiver is configured to send and/or receive signals from another transmitter-receiver mounted in the removable power tool **102** and/or the external accessory. The transmitter-receiver is configured to transmit one or more control signals for controlling the removable power tool **102** and/or the external accessory. In this way, the control of the removable power tool **102** can be remotely actuated by the power tool carriage **108** via transmission of actuation signals. In some examples the transmitter-receiver mounted on the power tool carriage **108** is configured to transmit actuation signals directly to a transmitter-receiver mounted in the external accessory. Additionally or alternatively transmitter-receiver mounted on the power tool carriage **108** is configured to transmit actuation signals indirectly to a transmitter-receiver mounted in the external accessory via a transmitter-receiver mounted in the removable power tool **102**. In this way, the switch **902** can send a signal to switch off the vacuum pump **904** and/or the water pump **1000**.

Water Pump

(97) A problem with some removable power tools **102** e.g. diamond core drills is that they need suitable lubrication management during operation. The power tool stand **100** as shown in the Figs. comprises a suitable lubrication management which will now be discussed in more detail.

(98) As mentioned above, the power tool carriage **108** comprises at least one duct **222** configured to convey a fluid. The at least one duct **222** is best shown in FIG. **2**. The at least one duct **222** in some examples is configured to supply a lubrication fluid such as water to the removable power tool **102**. Accordingly, the at least one duct **222** is in fluid communication with a lubrication fluid supply e.g. a water supply such as a water pump **1000** or a water tap (not shown). This allows the at least one duct **222** to supply water to the removable power tool **102** during use.

(99) In some other examples, the at least one duct **222** is in fluid communication with an air supply or a vacuum source (not shown). Accordingly, the at least one duct **222** is in fluid communication

with an air supply or a vacuum source e.g. a compressed air supply (not shown) or a vacuum pump (not shown) configured to extract dirt and debris. This allows the at least one duct **222** to extract debris from the removeable power tool **102**.

(100) The at least one duct **222** can be optionally used for conveying a fluid e.g. a lubrication fluid such as water towards the removeable power tool **102** or conveying a fluid e.g. an airflow away from the removeable power tool **102**. Hereinafter, the at least one duct **222** will be referred to as a water duct **222**, but this could also be an air duct as previously discussed.

(101) As shown in FIG. 2 the water duct **222** is mounted on the outside of the power tool carriage **108**. The water duct **222** extends along the surface of the power tool carriage **108**. Although not shown, in some examples, the water duct **222** can be secured to the power tool carriage **108** with clips, clamps, or adhesive or any other suitable means. In some alternative examples which are shown in e.g. FIG. 4, the water duct **222** is mounted inside the power tool carriage **108**.

(102) The water duct **222** comprises a duct inlet **404** in fluid communication with the water supply. In some examples, the duct inlet **404** comprises a toolless connector for connecting to a water hose. The toolless connector allows for a snap fit connection between the duct inlet **404** and a water hose (not shown). The water duct **222** is also in fluid communication with a fluid connector **224**. The water duct **222** can supply water to the moveable fluid connector **224**. The fluid connector **224** is optionally moveable between a first position and a second position. FIG. 2. shows the moveable fluid connector **224** in the first position. In the second position the fluid connector **224** is engaged with the removeable power tool **102**.

(103) The fluid connector **224** is mounted a connector housing **226**. The connector housing **226** protects the fluid connector **224** and houses a fluid connection between the water duct **222** and the fluid connector **224**.

(104) Alternatively, or additionally the fluid connector **224** is connectable with an accessory such as a nozzle (not shown). The optional nozzle is arranged to eject a spray of water on the cutting tool **132** in order to cool and lubricate the cutting tool **132**. In some other examples, the fluid connector **224** is connectable to an optional adapter (not shown) for guiding the flow of water around the cutting tool **132** and/or the removeable power tool **102**.

(105) In some examples, the water duct **222** comprises a carriage valve (not shown) arranged to selectively adjust the flow of fluid to and/or from the portion of the removeable power tool **102**. The carriage valve is arranged to be adjustable by the user to selectively adjust the flow of water to the cutting tool **132**. The user can optionally fully open or fully close the carriage valve in order to actuate the flow of water through the water duct **222**. Furthermore, the carriage valve can optionally be positioned at any position between a fully closed and a fully opened state. This means that the user can adjust the flow rate by manipulating the carriage valve on the power tool carriage **108** depending on the amount of lubrication required during a cutting operation.

(106) As mentioned above, in some examples, the power tool stand **100** comprises a water pump **1000** as shown in FIG. 10. The water pump **1000** as shown in FIG. 10 is mounted inside the carriage housing **200** of the power tool carriage **108**. Alternatively, the water pump **1000** can be located outside of the power tool carriage **108**. The water pump **1000** can be mounted to the outside of the power tool carriage **108**. Alternatively, the water pump **1000** can be mounted to the base **104**.

(107) The water pump **1000** is electrically connected to the removeable battery **130** when the removeable battery **130** is mounted to the power tool carriage **108**. This means that the removeable battery **130** is configured to power to the water pump **1000**. In some examples the power tool carriage **108** comprises a water pump control switch **1002** connected in series between the water pump **1000** and the removable battery **130**. The user can then manually turn on the water pump **1000** when needed by actuating the water pump control switch **1002**. In some examples, the controller **1100** (best shown in FIG. 11 or 14) can selectively control actuation of the water pump **1000**. For example, the controller **1100** can issue a control signal to the water pump **1000** to actuate the water pump **1000** when the user actuates the power trigger switch **308** on the removeable power

tool **102**.

(108) The water pump **1000** is in fluid communication with a water reservoir **1004**. The water reservoir **1004** is optionally a bucket of water or a water tank. The water pump **1000** can be connected to the water reservoir **1004** via a water hose **1006**. The water reservoir **1004** is unpressurised. The water pump **1000** can be any suitable pump configured to draw water from the water reservoir **1004** and convey the water to the water duct **222**.

(109) In some examples, the removeable battery **130** is configured to be in electrical connection with both the removeable power tool **102** and the water pump **100** when the removeable battery **130** and the removeable power tool **102** are mounted to the power tool carriage **108**.

(110) In some other examples, the removeable battery **130** mounted to power tool carriage **108** is in electrical connection with the water pump **1000** but not the removeable power tool **102**. In this case, the power tool carriage **108** comprises a power tool interface which mechanically couples the removeable power tool **102** to the power tool carriage **108**. However, when the removeable power tool **102** and the removeable battery **130** are mounted on the power tool carriage **108**, the removeable power tool **102** and the removeable battery **130** are in different electrical circuits. This means that the removable battery **130** is only in electrical connection with the water pump **1000**. In this example, the removeable power tool **102** is powered with an alternate power source e.g. another removeable battery or a mains power supply etc.

(111) In another example, the removeable power tool **102** is electrically connected to a first removeable battery **130** and the water pump **1000** is electrically connected to a second removeable battery **800**. Similarly the first removable battery **130** and the second removeable battery **800** are not electrically connected together and are in different electrical circuits. The first removeable battery **130** and the second removeable battery **800** will be discussed in more detail below.

Dual Battery and Battery Control

(112) As mentioned above, in some examples the power tool stand **100** is configured to mechanically and electrically connect to a plurality of batteries **130**, **800**. For example, a first removeable battery **130** and a second removeable battery **800** can be mounted on the power tool carriage **108** side by side as shown in FIG. **8**. The power tool carriage **108** comprises a first battery interface **802** and a second battery interface **804** configured to respectively electrically and mechanically connect the first and second removeable batteries **130**, **800** to the power tool carriage **108**. The first battery interface **802** and the second battery interface **804** are identical to the battery interface **206** discussed above and shown in e.g. FIG. **2**. Alternatively, any number of removeable batteries **130** can be mounted on a similar number of battery interfaces **204**.

(113) Any of the other examples can be used together with the arrangement discussed in reference to FIG. **8**. For example, the examples using the water pump **1000** as shown in FIG. **10** or the examples using the vacuum pump **904** as shown in FIG. **9** can be used together with the examples discussed in reference to FIG. **8**.

(114) In some examples, each of the first removeable battery **130** and the second removeable battery **800** are mounted to the power tool carriage **108** with the battery adaptor **504** as shown in FIGS. **5** and **7**. In other words, each of the first battery adaptor **806** and the second battery adaptor **808** as shown in FIG. **8** is the same as the battery adaptor **504** as discussed in reference to FIGS. **5** and **7**.

(115) Examples described with reference to FIGS. **8**, **14** and **15** discuss the control of the first battery **130** and the second battery **800**. FIG. **14** shows a schematic view of the power tool stand **100** with the removeable power tool **102**, the first battery **130** and the second battery **800** mounted to the power tool carriage **108**. FIG. **15** shows a flow diagram of operation of a power tool stand **100**.

(116) The first battery **130** and the second battery **800** can be each a single battery pack. In other examples, each of the first battery **130** and the second battery **800** can be each a group of battery packs or a group of battery cells. In some examples there can be other battery packs. There can be

any number of other battery packs as required. The example discussed below will only have a first battery **130** and a second battery **800** however the control functionality can be applied to any number of battery packs connected to the circuit **1400** of the power tool stand **100**. The first and second batteries **130**, **800** are identical to the removeable battery **130** as shown in FIG. 1 or alternatively identical to the high capacity removeable battery **502** as shown in FIG. 5.

(117) The first and second batteries **130**, **800** are electrically connectable to the removeable power tool **102** when mounted on the power tool carriage **108**. The first and second batteries **130**, **800** can be connected together to the removeable power tool **102** in series or parallel. Alternatively, first and second batteries **130**, **800** are connectable separately to the removeable power tool **102**. In other words, only one of the first and second batteries **130**, **800** is connected to the removeable power tool **102** at any one time.

(118) In this example, either a controller **1100** or a user is configured to selectively connect the first battery **130** and/or the second battery **800** to the removeable power tool **102**. A user can selectively connect the first battery **130** and/or the second battery **800** to the removeable power tool **102** by pressing a user actuated switch **1402** on the power tool stand **100**. The user actuated switch **1402** may change the connection between the first battery **130** to the removeable power tool **102** to a connection between the second battery **800** and the removeable power tool **102**.

(119) In some examples, the controller **1100** can switch the connection of first and second batteries **130**, **800** to the removeable power tool **102** between a series connection of the first and second batteries **130**, **800**, a parallel connection of the first and second batteries **130**, **800**, a single connection of the first battery **130** and a single connection of the second battery **800**.

(120) The controller **1100** is connected to the first battery **130** and the second battery **800**. The controller **1100** is configured to issue control instructions to the first and second batteries **130**, **800**. In some examples, the controller **1100** is a separate battery controller configured to control the connections of the first and second batteries **130**, **800** to the removeable power tool **102**.

(121) However, in other examples the controller **1100** is mounted within the power tool carriage **108** and is configured to control the connections of the first and second batteries **130**, **800**. Accordingly the controller **1100** is configured to selectively connect the first battery **130** and/or the second battery **800** to the removeable power tool **102**.

(122) The controller **1100** is configured to receive signals relating to parameters and/or variables for the first and second batteries **130**, **800** as shown in step **1500**. In a first example, the controller **1100** optionally receives a manual input from a user. The manual input from the user is received from a user actuated battery switch **1402** mounted on the power tool carriage **108**. For example, the user actuated battery switch **1402** can be positioned next to the water pump control switch **1002** as shown in FIG. 10. When the controller **1100** receives the signal from the user actuated switch **1402**, the controller **1100** determines a status of the first and second batteries **130**, **800** as shown in step **1502**. For example, the actuation of the battery switch **1402** indicates to the controller **1100** that the first battery **130** connected to the removeable power tool **102** has a low charge status.

(123) The user may actuate the battery switch **1402** in response to the user seeing that the charge in the first battery **130** is depleted. The user may determine the charge status from the first battery **130** itself. For example, the first battery **130** may comprise charge indication LEDs (not shown). In another example, a display (e.g., a row of LEDs or a screen) may be provided on the power tool carriage **108** (or any other part of the power tool stand **100** that is in the line of sight of the user from the operating position) that displays a state of charge for each of the first and second batteries **130**, **800**. The display may be electrically connected to first and second batteries **130**, **800** directly or via the controller **1100**. In another example, the display may wirelessly communicate (e.g., via Bluetooth) with the controller **1100** to receive state of charge information wirelessly.

(124) Once the controller **1100** determines that the user has actuated the battery switch **1402**, the controller **1100** determines that a change in status of the first battery **130** has occurred as shown in step **1504**. This means that there is a required change connection status of the current battery e.g.

the first battery **130** connected to the removeable power tool **102**. The controller **1100** then issues a control instruction to the circuit **1400** to connect the second battery **800** to the removeable power tool **102** as shown in step **1506**. At the same time the controller **1100** issues another control instruction to disconnect the first battery **130** from the removeable power tool **102**. In this way the controller **1100** separately connects the first or second batteries **130, 800** to the removeable power tool **102** as shown in step **1508**. Once the second battery **800** is connected to the removable power tool **102**, the user can continue operating the removable power tool **102**.

(125) In another example, there is optionally no manual input from the user via the battery switch **1402**. Instead the controller **1100** selectively controls the connection of the first battery **130** and/or the second battery **800** automatically. For example, the controller **1100** can switch the connection to the removeable power tool **102** between the first and second batteries **130, 800** in dependence of the operational status of the first and second batteries **130, 800**.

(126) In some examples, in step **1500** the controller **1100** is configured to receive operational status information from the first and second batteries **130, 800**. The operational status information can comprise one or more of the following battery voltage, battery capacity, battery temperature, battery discharge rate, battery current or any other battery parameter or status information. In some examples, the controller **1100** is connected to a voltage sensor **1404** and a current sensor **1406** to respectively send voltage signals and current signals to the controller **1100**.

(127) Once the controller **1100** has received the signal from the first and second batteries **130, 800**, the controller **1100** determines the current status information of the first and second batteries **130, 800** in step **1502**. For example, the controller **1100** determines that the first battery **130** which is connected to the removeable power tool **102** has an instantaneous operating temperature, an instantaneous operating voltage and/or an instantaneous remaining battery capacity.

(128) The controller **1100** then compares the determined status information of the first and second batteries **130, 800** to predetermined threshold values as shown in step **1510**. The predetermined threshold values may optionally comprise operating voltage ranges, operating battery capacity and operating battery temperature. In some examples, the predetermined threshold values are a look up table stored in memory of the controller **1100**.

(129) For example, in step **1510** the controller **1100** determines that the battery capacity of the first battery **130** connected to the removeable power tool **102** has dropped below a threshold of e.g. 30%, 25%, 20%, 15%, 10%, or 5% of the total battery capacity of the first battery **130**.

(130) Alternatively, in step **1510** the controller **1100** determines that the operating temperature of the first battery **130** connected to the removeable power tool **102** has exceeded an operating temperature threshold e.g. 60° C., 55° C., 50° C., 45° C., 40° C. etc.

(131) Once the controller **1100** has determined that the first battery **130** is operating outside the normal operating parameters in step **1510**, the controller **1100** determines similarly in step **1504** that the operational status of the first battery **130** has changed. Accordingly, the controller **1100** issues a control instruction in step **1506** to the circuit **1400** to connect the second battery **800** to the removeable power tool **102** and disconnect the first battery **130** from the removeable power tool **102** as shown in step **1508**.

(132) If the controller **1100** determines that the first battery **130** is operating normally in step **1504** and there has not been a change in the operational status of the first battery **130**, the controller **1100** does not issue a control instruction to change the connection to the removeable power tool **102** between the first and second batteries **130, 800**. Instead, the controller **1100** returns to step **1500** continues to receive signals relating to parameters and/or variables relating to the first and second batteries **130, 800**.

(133) Optionally, the controller **1100** can issue a control instruction to the circuit **1400** to connect the first and second batteries **130, 800** in parallel as shown in step **1512** or in series as shown in step **1514**. This can provide a different voltage and current to the removeable power tool **102** if required.

(134) As mentioned above, in some examples the first and second batteries **130, 800**, comprise battery circuits comprising a battery controller. The controller **1100** in some examples is configured to receive operational data from the battery controller as shown in step **1500**. In some examples, the operational data is one or more error signals e.g. an error code indicating a fault or malfunction of the first or second batteries.

(135) In this case in the controller **1100** determines a fault status of the first battery **130** in step **1502**. The controller **1100** determines that the first battery **130** has changed from normally operating to malfunctioning and needs to be disconnected from the removeable power tool **102**. The controller **1100** then connects the second battery **800** to the removeable power tool **102** and disconnects the first battery **130** as previously discussed.

(136) In another example, the controller **1100** is optionally configured to optimizing the voltage provided to the removeable power tool **102** by selectively connecting the first and second batteries **130, 800** to the removeable power tool **102**. In this way, the controller **1100** is configured to optimize the voltage provided to the removeable power tool **102** as the first and second batteries **130, 800** discharge.

(137) The controller **1100** is optionally configured to determine a removable power tool **102** parameter e.g. a motor parameter in step **1516**. The controller **1100** determines that the motor of the removeable power tool **102** may be rated for a first voltage, e.g., 40V. The controller **1100** in some examples determines the motor voltage rating based on information stored in memory of the controller **1100**. The controller **1100** may determine that the first and second batteries **130, 800**, at full charge, provide a second voltage. In some examples, the second voltage is higher than the first voltage e.g., 60V.

(138) In some cases, the controller **1100** selectively connects the first and second batteries **130, 800** to the removeable power tool **102**. This means that the controller **1100** does not need to use pulse width modulation (PWM) at a duty cycle to provide the lower first voltage to the removeable power tool **102** from e.g., the first battery **130** at the higher second voltage.

(139) This can be advantageous because the controller **1100** can avoid the inherent losses caused by PWM switching. In this case, the controller **1100** can selectively connect the first and second batteries **130, 800** to the removeable power tool **102** with a 100% duty cycle.

(140) In some examples, in steps **1502, 1504** the controller **1100** determines the current battery capacity of the first battery **130** and the second battery **800**. The controller **1100** may determine that the first battery **130** and the second battery **800** may have a different state of charge. Accordingly, the controller **1100** determines that the first battery **130** and the second battery **800** have different instantaneous voltages.

(141) After the controller **1100** determines the motor voltage rating in step **1516**, the controller **1100** determines which of the first and second batteries **130, 800** has an instantaneous voltage closer to the motor voltage rating. The controller **1100** in step **1506** connects to either the first battery **130**, or the second battery **800** in dependence of which instantaneous voltage of the first and second batteries **130, 800** is closer to the motor voltage rating. In this way, the controller **1100** can supply the correct voltage to the motor of the removeable power tool **102** whilst minimizing the PWM switching.

(142) In step **1506**, if the controller **1100** determines that the motor voltage rating is 40V and the first battery **130** has an instantaneous voltage of 60V and the second battery **800** has an instantaneous voltage of 45V, then the controller **1100** will connect the second battery **800** to the motor of the removeable power tool **102**. This is because the voltage of the second battery **800** is closer to the motor voltage rating. This will mean that it may be more efficient to run the motor of the removeable power tool **102** from the second battery **800** so that limited or no PWM switching is needed.

(143) In another example, controller **1100** determines that the motor of the removeable power tool **102** requires different voltages in dependence of the motor speed. Accordingly in step **1516**, the

controller **1100** determines the required motor speed. Information relating to the required motor speed in some examples is received by the controller **1100** in a signal. The signal may be generated by a user input e.g., a trigger switch or a speed dial. In step **1516**, the controller **1100** determines that a high motor speed or a low motor speed is required. The controller **1100** determines the high motor speed voltage or the low motor speed voltage. Accordingly, the controller **1100** determines the required motor voltage. The controller **1100** determines in step **1506** which of the voltages of the first or second batteries **130**, **800** most closely matches the required motor voltage.

(144) For example, the controller **1100** may determine that the required motor voltage for a low-speed rating is 40V and the first battery **130** has an instantaneous voltage of 60V and the second battery **800** has an instantaneous voltage of 40V. In this case, the controller **1100** determines in step **1506** that the second battery **800** voltage is closest to the required motor voltage. The controller **1100** then connects the second battery **800** to the motor of the removeable power tool **102**.

(145) Alternatively, the controller **1100** may determine that the required motor voltage for a high-speed rating is 60V and the first battery **130** has an instantaneous voltage of 60V and the second battery **800** has an instantaneous voltage of 40V. In this case, the controller **1100** determines in step **1506** that the first battery **130** voltage is closest to the required motor voltage. The controller **1100** then connects the first battery **130** to the motor of the removeable power tool **102**.

(146) Furthermore, in some examples, the controller **1100** is configured to connect the first and second batteries **130**, **800** in series in step **1514**, or in parallel in step **1512**, in order to generate a voltage supplied by the first and second batteries **130**, **800** as close as possible to match the required motor voltage.

Vacuum Seal

(147) Another example will now be discussed in reference to FIGS. **9**, **11**, **12**, **13** and **14**. FIG. **9** shows a side view of the power tool stand **100**. FIG. **11** shows a schematic view of the power tool stand **100**. FIG. **12** shows a graph of pressure of a workpiece surface seal **906** time. FIG. **13** shows a flow diagram of operation of the power tool stand **100**.

(148) FIG. **14** shows a schematic view of the power tool stand **100**.

(149) The power tool stand **100** as shown in FIG. **9** is the same as discussed in reference to the FIGS. **1** to **8** except that the base **104** is modified. The base **104** comprises a work surface seal **906** mounted in the base **104** and configured to engage the workpiece surface **920**. The work surface seal **906** is configured to maintain a vacuum seal against the workpiece surface **920**. This means that the work surface seal **906** provides a suction force for fixing the power tool stand **100** against the workpiece surface **920**. The workpiece surface **920** in some examples (not shown in FIG. **9**) is a vertical surface such as a wall. This means that the work surface seal **906** allows the power tool stand **100** to be more easily or better fixed to the vertical surface.

(150) Whilst FIG. **9** shows a single work surface seal **906**, in other examples there can be a plurality of work surface seals **906** mounted on the base **104**.

(151) In some examples, the work surface seal **906** generates a sufficiently large suction force that only the work surface seal **906** is needed to fix the power tool stand **100** to the workpiece surface **920**. In other examples, the work surface seal **906** is used together with other fixing attachments e.g. the fixing bolts **120** as shown in FIG. **1**.

(152) The work surface seal **906** is in fluid communication with a vacuum source **904**. In some examples the vacuum source **904** is a vacuum pump **904** connected to the work surface seal **906** via a hose **908**. Whilst FIG. **9** shows the vacuum pump **904** remote from the power tool stand **100**, in some other examples, the vacuum pump **904** is mounted on the power tool carriage **108** or the base **104**. The vacuum pump **904** as shown in FIG. **9** is optionally mounted in a vacuum pump housing which is mounted to the base **104**. The vacuum pump housing can be fixed to base **104** via threaded fasteners. One or more suction conduits or hoses are in fluid communication between the vacuum pump **904** and the work surface seal **906**. The one or more suction conduits or hoses can extend through the base **104** to the work surface seal **906** as schematically shown in FIG. **9**.

(153) Alternatively, the vacuum pump **904** is mounted within the power tool carriage **108**. The vacuum pump **904** comprises a motor fan assembly **910** configured to generate a negative pressure airflow. When the vacuum pump **904** is actuated, the vacuum pump **904** is configured to generate a partial pressure or vacuum pressure in the work surface seal **906** which causes a suction force on the work surface seal **906** against the workpiece surface **920**.

(154) Any suitable vacuum source **904** means can be used to generate to the negative pressure air flow to cause the suction force. For example, the arrangement as shown in FIG. **9** shows a vacuum source **904** connected via the hose **908** to the work surface seal **906**. Alternatively, the negative pressure air flow can be generated next to the work surface seal **906** in the base **104**. For example, a venturi nozzle (not shown) mounted next to and in fluid communication with the work surface seal **906** can be connected to a source of compressed air. The venturi nozzle can generate the negative pressure air flow when the source of compressed air flows into the venturi nozzle. The vacuum pump **904** can be manually actuated by a user operated vacuum pump control switch **1106**. This means that the control of the vacuum pump **904** can be achieved manually and without the need of a controller **1100** or a pressure sensor **912** as shown in FIG. **9**. In some examples, once the user manually actuates the vacuum pump **904**, the vacuum pump **904** is configured to generate a vacuum pressure at a predetermined pressure e.g. a pressure set point. This means that the suction force can be applied at the work surface seal **906** without automatic control.

(155) The vacuum pump **904** is optionally electrically connected to the removeable battery **130** when the removeable battery **130** is mounted to the power tool carriage **108**. This means that the removeable battery **130** is configured to power to the vacuum pump **904**. In some examples the power tool carriage **108** comprises a vacuum pump control switch **1106** connected in series between the vacuum pump **904** and the removable battery **130**. The user can then manually turn on the vacuum pump **904** when needed by actuating the vacuum pump control switch **1106**. In some other examples, the vacuum pump **904** is alternatively optionally electrically connected to a separate independent battery (not shown) mounted to the vacuum pump **904** or the power tool stand **100**. In this case, there are two removable batteries **130** mounted to the power tool stand which are each part of a separate electrical circuit e.g. an electrical circuit for the removeable power tool **102** when mounted on the power tool stand **100** and another electrical circuit for the vacuum pump **904**.

(156) Additionally or alternatively, the vacuum pump **904** can be optionally controlled by a controller **1100** and vacuum source **904** is connected to a controller **1100**. Alternatively, this means that the vacuum pump **904** can be controlled without manual input from the user as mentioned above. The controller **1100** is configured to issue control instructions to the vacuum source **904** and/or the removeable power tool **102**. For example, the controller **1100** is configured to actuate and control the speed and direction of the motor fan assembly **910**. The controller **1100** is also configured to control the removeable power tool **102** in response to one or more detected conditions in the work surface seal **906**. The controller **1100** is connected to the removeable power tool **102** via a first data connection **914**. The controller **1100** as shown in FIG. **9** is external to the power tool stand **100**. However, in some other examples, the controller **1100** is mounted within the power tool carriage **108**.

(157) The controller **1100** is also optionally connected to at least one sensor **912** via a second data connection **916**. The at least one sensor **912** is a pressure sensor **912** configured to detect the pressure in the work surface seal **906**. The pressure sensor **912** is configured to send a pressure signal to the controller **1100**. The pressure sensor **912** can periodically detect the pressure and send a pressure signal to the controller **1100**. In other examples, the pressure sensor **912** constantly sends a pressure signal to the controller **1100**. In some examples, the pressure sensor **912** is configured to detect the vacuum pressure at the work surface seal **906**.

(158) In some other examples, the controller **1100** is configured to receive the raw pressure signal from the pressure sensor **912** and determine the vacuum pressure at the work surface seal **906**.

(159) The controller **1100** in some examples is optionally connected to a vacuum pump voltage

sensor **1102** and a vacuum pump current sensor **1104** which are configured to respectively send voltage and current signals relating to the motor fan assembly **910** of the vacuum source **904**. This means that the controller **1100** can optionally determine the load on the vacuum source **904** and determine whether there is a fault with the vacuum source **904** or whether an adequate seal has been made between the work surface seal **906** and the workpiece surface **920**.

(160) Operation of the work surface seal **906** will now be discussed in further detail with respect to the FIG. **13**.

(161) The controller **1100** receives a pressure signal from the pressure sensor **912** in step **1300**. The controller **1100** then determines whether the received pressure signal is below a first vacuum pressure threshold **1200** as shown in step **1302**. At $T=T_1$ as shown in FIG. **12** the controller **1100** determines that the received pressure signal is at a first vacuum pressure signal. It should be noted that the graph in FIG. **12** shows the vacuum pressure at the work surface seal **906** over time. This means a higher vacuum pressure indicates a lower absolute pressure and a lower vacuum pressure indicates a higher absolute pressure. In other words as the vacuum pressure tends to zero in FIG. **12**, the absolute pressure in the work surface seal **906** tends to atmospheric pressure.

(162) The first vacuum pressure threshold **1200** is a threshold that indicates that the seal between the work surface seal **906** and the workpiece surface **920** is not perfect. This means that there is a suction force being exerted by the work surface seal **906**, but this is not a maximum suction force that can be exerted by the work surface seal **906**.

(163) When the vacuum pressure in the work surface seal **906** is between the first vacuum pressure threshold **1200** and a second vacuum pressure threshold **1202**, the power tool stand **100** and removeable power tool **102** are operable. However, the user should be notified that suction force may not be sufficient to safely operate the removeable power tool **102** on the power tool stand **100**. The second vacuum pressure threshold **1202** is lower than the first vacuum pressure threshold **1200**. This means that a pressure in the work surface seal **906** at the second vacuum pressure threshold **1202** has a worse seal between the work surface seal **906** and the workpiece surface **920** than when the pressure in the work surface seal **906** is at the first vacuum pressure threshold **1200**.

(164) Since the controller **1100** determines that the pressure is above the first vacuum pressure threshold **1200**, the controller **1100** returns to step **1300** and continues monitoring the pressure in the work surface seal **906**.

(165) At time $T=T_2$, the controller **1100** determines that the pressure in the work surface seal **906** has dropped below the first vacuum pressure threshold **1200**. This means that the vacuum pressure in the work surface seal **906** is dropping. The controller **1100** then determines whether the pressure in the work surface seal **906** has dropped below a second vacuum pressure threshold **1202** in step **1304**. At $T=T_2$, the controller **1100** determines that the pressure is still above the second vacuum pressure threshold **1202**.

(166) The controller **1100** then issues an alert to the user in step **1306**. The alert can be displayed on an LCD display (not shown) of the removeable power tool **102** or on the power tool stand **100**. Alternatively, the alert can be illuminating an LED on the removeable power tool **102** or the power tool stand **100**. Alternatively, the alert can be an audible signal or a haptic signal.

(167) If the controller **1100** determines whether the pressure in the work surface seal **906** has dropped below a second vacuum pressure threshold **1202** in step **1304**, e.g. at $T=T_3$, the controller **1100** can issue a control instruction to stop the removeable power tool **102** as shown in step **1308**. The controller **1100** may actuate a switch to cut the power to the removeable power tool **102** in response to determining that the vacuum pressure at the work surface seal **906** has fallen below the second vacuum pressure threshold **1202**. In this case, the controller **1100** may stop the vacuum source **904** as well. Accordingly the vacuum pressure falls to zero at $T=T_4$.

(168) This means that the user can be protected from the power tool stand **100** detaching from the workpiece surface **920** and falling on the user.

(169) In some examples, the controller **1100** in step **1310** may optionally issue a control instruction

to the vacuum source **904** following a determination that the vacuum pressure is falling in the work surface seal **906** at $T=T5$. The controller **1100** issues a control instruction to increase the speed of the motor fan assembly **910** and increase the negative pressure air flow. This can increase the vacuum pressure in the work surface seal **906** such that the vacuum pressure exceeds the first vacuum pressure threshold **1200** at $T=T6$. The controller **1100** can control the pressure in the work surface seal **906** with a feedback loop from the pressure sensor **912** and controlling the vacuum source **904** as shown in FIG. **13**.

(170) Advantageously, by using the pressure sensor **912** which detects the vacuum pressure in the work surface seal **906**, the controller **1100** can directly determine whether the vacuum seal against the workpiece surface **920** has deteriorated more accurately. This means the controller **1100** can inform the user more accurately and safely in case there is an issue with the work surface seal **906**.

(171) In another example, two or more examples or features are combined. Features of one example can be combined with features of other examples e.g. any of the features described above in one example or shown in one or more Figs can be combined with one or more other features described above or shown in one or more Figs. Examples of the present disclosure have been discussed with particular reference to the examples illustrated. However it will be appreciated that variations and modifications may be made to the examples described within the scope of the disclosure.

Claims

1. A power tool stand for a removeable power tool having a first battery interface and a removeable battery having a first tool interface, the power tool stand comprising: a base; a projecting frame connected to the base; and a power tool carriage moveably mounted on the projecting frame, the power tool carriage comprising a second battery interface and a second tool interface; wherein the second battery interface is compatible with the first tool interface and the second battery interface is arranged to electrically and mechanically connect together with the first tool interface; and the second tool interface is compatible with the first battery interface and the second tool interface is arranged to electrically and mechanically connect together with first battery interface; and the first battery interface and the first tool interface are incompatible.
2. A power tool stand according to claim 1, wherein the power tool carriage comprises a battery adapter comprising the second battery interface.
3. A power tool stand according to claim 2, wherein the battery adapter is mountable to the power tool carriage.
4. A power tool stand according to claim 3, wherein the battery adaptor is removably mountable to the power tool carriage.
5. A power tool stand according to claim 2, wherein the battery adapter comprises a carriage interface and the power tool carriage comprises an adaptor interface and adaptor interface is compatible with the carriage interface and the adaptor interface is arranged to electrically and mechanically connect together with carriage interface.
6. A power tool stand according to claim 5, wherein the carriage interface is the same as the second tool interface and the adaptor interface is the same as the first battery interface.
7. A power tool stand according to claim 1, wherein the second battery interface is configured to mechanically and electrically connect to the removable battery, and the removable battery has a voltage of at least 60V.
8. A power tool stand according to claim 1, wherein the power tool stand comprises a rack and pinion mechanism arranged to move the power tool carriage relative to the projecting frame.
9. A power tool stand according to claim 1, wherein the power tool carriage comprises a handle coupled to the rack and pinion mechanism.
10. A power tool stand according to claim 1, wherein the second battery interface is mounted on a

first side of the power tool carriage and the second tool interface is mounted on a second side of the power tool carriage.

11. A power tool stand according to claim 1, wherein the second tool interface projects over the base.

12. A power tool stand according to claim 1, wherein the power tool stand comprises a plurality of the second battery interfaces arranged to electrically connect with a plurality of removeable batteries.

13. A power tool stand according to claim 1, wherein the power tool carriage comprises a latch recess for engaging with a latch mechanism mounted on the removeable battery.

14. A power tool stand according to claim 1, wherein the power tool carriage comprises an air duct connectable to a dust extractor.

15. A power tool stand according to claim 1, wherein the power tool carriage comprises a water duct connectable to a water supply.

16. A power tool stand according to claim 1, wherein the second battery interface is electrically connected to the second tool interface within the housing of the power tool carriage.

17. A power tool stand system comprising: a power tool stand, the power tool stand comprising: a base; a projecting frame connected to the base; and a power tool carriage moveably mounted on the projecting frame. the power tool carriage comprising a second battery interface and a second tool interface; a removeable power tool having a first battery interface, wherein the first battery interface is connectable to the second tool interface of the power tool stand; and a first removable battery having a first tool interface, wherein the first tool interface is connectable to the second battery interface of the power tool stand.

18. A power tool stand system according to claim 17, further comprising a second battery having a third tool interface compatible with the first battery interface.

19. A power tool stand, comprising: a base; a projecting frame; and a power tool carriage configured to support a removable power tool and to translate relative to the projecting frame, the power tool carriage comprising a second battery interface and a second tool interface; wherein the second battery interface is configured to electrically and mechanically connect together with a first tool interface of a removable battery pack; wherein the second tool interface is configured to electrically and mechanically connect together with a first battery interface of the removable power tool; and wherein the first battery interface and the first tool interface are incompatible.

20. The power tool stand of claim 19, wherein the second battery interface is at a first side of the power tool carriage and the second tool interface is at a second side of the power tool carriage.
